Ling Chang

lc203@rice.edu San Francisco https://www.linkedin.com/in/lingchangtw

EDUCATION

Rice University Houston, TX
Master of Computational Science and Engineering *Aug. 2025 - Expected Dec. 2026*

- Coursework: Database System, Machine Learning, Computer System and Architecture, IoT Programming

The University of Texas at Dallas Richardson, TX
Graduate Coursework in Information Technology and Management *Aug. 2024 - Jul. 2025*

- Coursework: Object-Oriented Programming, Advanced Statistics & Data Science, Causal Analytics, Big Data

SKILLS

- **Programming Languages:** Python, Kotlin, C++, JavaScript, SQL, PHP, C#
- **AI & LLMs:** Large Language Model (LLM) Agents, Model Context Protocol (MCP), Knowledge Graphs
- **Cloud & Developer Tools:** AWS, GCP, React.js, HTML/CSS, Docker, Git, CI/CD, Linux, Shell Scripting
- **Database Systems:** PostgreSQL, MySQL, MongoDB, Neo4j, Apache Spark, Hadoop

EXPERIENCE

Intuit Inc. New York City, NY
Full-Stack Software Engineer Intern *May 2026 - Aug. 2026*

- Drove a monolith-to-cloud data migration to reduce operating costs and modernize the platform, moving a customer-facing audience segmentation feature from a legacy monolith database to a cloud-based Customer Data Platform (CDP).
- Built the first integration between a backend microservice and the CDP using Kotlin, PHP, and MySQL. Delivered the first e-commerce entity end-to-end, extended it to a second, and established a reusable migration pattern for the remaining entities.
- Owned the production roll-out using shadow validation and feature flags to verify data parity and reduce deployment risk, and identified a data inconsistency in the legacy system that was resolved to improve data accuracy.

The University of Texas MD Anderson Cancer Center Houston, TX
Graduate Research Assistant *Jan. 2026 - May 2026*

- Built an internal wiki search platform for the lab's oncology datasets, adopted by 10 researchers to prevent duplicate analyses and speed up data discovery.
- Implemented and evaluated LLM-based predictor pipelines for high-dimensional sequential biomedical data.

SELECTED PROJECTS

KG2Prompt: Knowledge Graph Platform for Agentic LLM Prompt Construction

- Developed a full-stack platform for KG-guided LLM prompt optimization and interactive KG visualization.
- Built an LLM agent system integrating biomedical KG database for dynamic prompt generation.
- Built a Neo4j-backed architecture for prompt generation, reducing response latency by 38% for physician users.

Flight Reservation Database System Design

- Designed and implemented a PostgreSQL database with ERDs, relational schema, and integrity constraints to support flight reservation workflows.
- Optimized database performance through 3NF normalization and complex SQL queries, reducing data redundancy by 22% and improving query efficiency by 31%.

PUBLICATION

[NeurIPS' 25 Workshop] Y.N. Chuang, L. Yu, G. Wang, L. Zhang, **L. Chang**, et. al., and X. Hu. "Confident or Seek Stronger: Exploring Uncertainty-Based On-device LLM Routing"